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December 15, 1972
SUPERSEDING
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April 11, 1967

FEDERAL SPECIFICATION

GAGES, TIRE PRESSURE, SELF-CONTAINED AND INFLATOR-GAGE, PNEUMATIC TIRE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers direct-reading tire pressure and inflator gages, calibrated in pounds per square inch (p.s.i.).

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Tire pressure and inflator gages covered by this specification shall be of the following types, classes, and styles as specified (see 6.2):

Type I - Precision gage.

Type II - General service gage.
  Class A - Heavy duty.

    Styles:
      1 - Short extension, single-chuck foot.
      2 - Long extension, dual-chuck foot.

  Class B - Light duty, single-chuck foot.

    Styles:
      1 - Low pressure.
      2 - High pressure
      3 - Low pressure, air-liquid.
      4 - Extra low pressure.

Type III - Inflator gage, internal indicating.

    Styles:
      1 - Single-chuck foot.
      2 - Dual-chuck foot.
      3 - Dual-chuck foot, standard and large bore.

Type IV - Inflator gage, external indicating.

    Styles:
      1 - Single-chuck foot.
      2 - Dual-chuck foot.
      3 - Dual-chuck foot, standard and large bore.

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

QQ-P-416 - Plating, Cadmium (Electro-deposited).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes: Paperboard, Metal-Edged and Components.
PPP-B-676 - Boxes, Setup.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, and Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-C-4109 - Coupling Assembly, Low Pressure, Air Hose, Quick-Disconnect.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods.
MS-33586 - Metals, Definition of Dissimilar.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS), Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

Uniform Classification Committee; Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Tariff Publishing Officer, Room 1106, 222 South Rineside Plaza, Chicago, Illinois 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish one complete gage for first article inspection. The first article may be either a pre-production model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Materials. Materials not specifically designated shall be compatible with the intended use and requirements specified herein. Synthetic materials shall not burn at a rate exceeding 2 inches per minute following an ignition. Materials having a cellulose nitrate base are not acceptable. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS 33586.

3.2.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Strength and other essential physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein.

3.3 Construction. The overall construction of the gages shall be free from any characteristics or defects that might render the gages unsuitable or inefficient for the intended purpose. Types I and II gages shall withstand a 25 percent plus or minus 5 percent overload air pressure, without damage, for a minimum of 10 minutes.

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Types III and IV gages shall withstand an air pressure of 175 p.s.i. for a minimum of 10 minutes. All parts subject to maintenance disassembly shall be accessible for adjustment, repair, and cleaning without the necessity of special tools. The gage cartridge for type III gages shall be assembled either by threaded plugs or swedging method to prevent the internal mechanism from becoming disassembled during installation. Type II, class A gages shall withstand a force of 125 pounds applied at the center of the gage without permanent deformation and shall withstand penetration of a 0.0625-inch radius test pin when a force of 125 pounds is applied at any point on the barrel or stem. Slight deformation or marking not affecting serviceability or appearance is acceptable.

3.3.1 Endurance. The gages shall be constructed as to withstand not less than 2,000 actual gaging operations for types I and II gages, and 10,000 actual gaging operations for types III and IV inflator gages at their maximum operating pressure range, without the necessity of recalibration, adjustment, or repair.

3.3.2 Safety devices. Each tire gage shall be equipped with safety devices where practical. Parts which are hazardous to the operator shall be guarded.

3.3.3 Shock. The tire gages covered by this specification shall withstand being dropped free fall, from a minimum height of 3 feet, to a concrete surface five times for types I and II gages and fifty times for types III and IV inflator gages.

3.3.4 Welding, brazing, and soldering. Welding, brazing, and soldering shall be employed only where these operations are required in the original design. Such processes shall not be employed as repair measures for defective parts.

3.3.5 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.6 Threads. All threads shall be in inch units of measurement and comply with NBS H28.

3.3.7 Interchangeability. All replaceable parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model can be replaced without requiring modification. Where practicable, all such parts shall be permanently and legibly marked with the manufacturer's part number in accordance with MIL-STD-130.

3.3.8 Metal finish. All functioning surfaces of the gages shall be smooth and free from tool marks and extraneous materials. All exposed sharp edges shall be broken with a radius or chamfer, as applicable.

3.3.9 Protective finish. Unless otherwise specified (see 6.2), all metal parts shall be plated, treated, or be of corrosion resistant material to withstand the corrosive action of salt, fog, and high relative humidity as encountered in general service. Recalibration devices shall be of rust-resisting materials, but not necessarily plated. Any cadmium plating shall be accomplished in accordance with the requirements of QQ-P-416.

3.4 Design. Tire pressure gages furnished under this specification shall be convenient to use and entirely adequate in every respect for the intended purpose. Specific design characteristics shall be as specified herein for each type, class, and style.

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3.4.1 Illustrations. The illustrations shown in Figures 1 through 8 herein are for the convenience of identification and are not intended to preclude the purchase of gages which are otherwise in accordance with the requirements of this specification.

3.4.2 Chuck foot. Unless otherwise specified (see 6.2), the chuck foot of all gages shall be offset at an angle to the centerline of the gage barrel. The chuck(s) shall be so constructed as to fit standard (0.302 - 32) threads used on tire valves, or when hereinafter defined as large bore, the chuck foot shall fit (0.482 - 26) valve threads used on off-road tires. When applied to the valve stem, the chuck(s) shall depress the valve core plunger permitting the free passage of air from the tire into the gage pressure chamber without loss of air or damage to the stem or core members. The chuck foot shall be made integral with the extension or connecting member, or rigidly attached thereto in such a manner as not to become loose or detached.

3.4.2.1 Single-chuck foot. This type foot shall be provided with a single chuck which shall be so constructed as to operate the gage when depressed on the valve stem. Type II gages shall also be provided with an external deflator pin.

3.4.2.2 Dual-chuck foot. This type of foot shall be provided with a dual opposed deflator pin. The dual-chuck foot assembly shall be so constructed as to operate the gage from one chuck and render the other chuck deflator pin inoperative. The deflator pin shall be hardened steel and shall be constructed so that distortion of the rubber sealing washer cannot partially or completely block the flow of air through the gage.

3.4.3 Types I and II gages. Gages of types I and II shall be composed primarily of a barrel with indicator bar and single or dual-chuck foot with extension, except that type II, class B shall have no extension.

3.4.3.1 Barrel. The barrel shall be so constructed as to fully enclose and shield the internal working mechanism. The inside of the barrel shall be corrosion resistant or treated to prevent corrosion along its entire length.

3.4.3.2 Indicator bar. The indicator bar shall be furnished in circular, triangular, or square cross section. The connection between the indicator bar and the internal actuating mechanism shall be such as to permit the indicator bar to remain at the indicated pressure after the source of pressure has been removed, except for type II, class B, styles 3 and 4. The graduation marks shall be stamped, etched, or otherwise set to a width and depth to provide a permanency at least equal to the life of the gage. The accuracy in graduation spacings shall be such as to meet with the requirements of each particular type and class. The numerals indicating the applicable graduation increment shall be a minimum 0.0625 inch in height. Both graduations and numerals shall be clear and well defined to facilitate reading of the gage under normal service conditions.

3.4.3.3 Type I precision gage. Type I precision gages shall have a low pressure reading of 10 pounds per square inch gage (p.s.i.g.) or less, and a high pressure reading of not less than 130 p.s.i.g. The indicator bar shall be graduated in 2 p.s.i. increments, or less, the entire length of the scale. The scale shall be marked with the appropriate numerical value at each 10 p.s.i. interval. The graduated length of the scale shall be not less than 2.5 inches nor more than 3.5 inches. The gage shall be 12 inches plus or minus 2 inches in overall length and shall be provided with an extension 3.5 to 4.5 inches in length having a standard bore dual-chuck foot. Each gage shall be provided with positive identification to avoid mistaking the precision gage and general purpose gage. Gages of this type shall be similar in design to Figure 1.

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3.4.3.3.1 Accuracy. Precision gages at plus 70° Fahrenheit (F.), and minus 20° F., and within 5 seconds after pressure has been applied, shall indicate the true pressure within the following limits:

Accuracy at plus 70° F., plus or minus 5° F. (room temperature), to be within plus or minus:

1 p.s.i. at 30 p.s.i.g.
1 p.s.i. at 50 p.s.i.g.
1.5 p.s.i. at 75 p.s.i.g.
2 p.s.i. at 120 p.s.i.g.

Accuracy at minus 20° F. to minus 25° F. to be within plus or minus:

1 p.s.i. at 30 p.s.i.g.
1.5 p.s.i. at 50 p.s.i.g.
2 p.s.i. at 75 p.s.i.g.
3 p.s.i. at 120 p.s.i.g.

3.4.3.4 Type II, general service gage. Type II general service gages shall be of two classes for general service usage, as specified herein.

3.4.3.4.1 Type II, class A heavy duty general service gage. Gages of this type and class shall be of heavy construction throughout to withstand rough usage as encountered in service stations and tire shops. The type II, class A gages shall have a low pressure reading of 20 p.s.i.g. or less, and a high pressure reading of not less than 130 p.s.i.g. The indicator bar shall be graduated in 2 p.s.i. increments or less up to at least 60 p.s.i.g. on the scale and in 5 p.s.i.g. increments or less for the remainder of the scale. The scale shall be marked with the appropriate numerical value at each ten-pound unit or less. The length of the graduated scale shall not be less than 2.875 inches.

3.4.3.4.1.1 Accuracy. Gages of this type shall indicate the true pressure at plus 70° F. and minus 20° F., and within 5 seconds after pressure has been applied, within the following limits:

Accuracy at plus 70° F. plus or minus 5° F. (room temperature), to be within plus or minus:

1.5 p.s.i. at 30 p.s.i.g.
1.5 p.s.i. at 50 p.s.i.g.
2 p.s.i. at 75 p.s.i.g.
3 p.s.i. at 120 p.s.i.g.

Accuracy at minus 20° F. to minus 25° F. to be within plus or minus:

1.5 p.s.i. at 30 p.s.i.g.
2 p.s.i. at 50 p.s.i.g.
3 p.s.i. at 75 p.s.i.g.
3 p.s.i. at 120 p.s.i.g.

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3.4.3.4.1.2 Type II, class A, style 1, short-extension single-chuck foot gage. Gages of this type, class, and style shall be 8.5 to 9.5 inches in overall length, and provided with an extension 1.5 to 2.5 inches in length having a standard bore single-chuck foot. These gages shall be similar in design to Figure 2.

3.4.3.4.1.3 Type II, class A, style 2, long extension dual-chuck foot gage. Unless otherwise specified (see 6.2), the gages of this type, class, and style shall be not less than 11 inches in overall length. Each gage shall have a standard bore dual-chuck foot and provisions for hanging up the gage. These gages shall be similar in design to Figure 3.

3.4.3.4.2 Type II, class B, light duty gage. Gages of this type and class shall be of four styles and shall be for light duty, individual usage. The type II, styles 1, 2, and 3, class B gages shall have a pocket clip, a standard bore single-chuck foot, and shall be 5 to 6 inches in overall length with indicator bar retracted. The type II, class B, style 4 gage shall be 4.5 to 7.5 inches in overall length with indicator bar retracted.

3.4.3.4.2.1 Accuracy. Type II, class B, styles 1 and 3 gages shall indicate the true pressure within plus or minus 1 p.s.i., at plus 70° F. plus or minus 5° F. temperature, at 40 p.s.i.g. pressure. Type II, class B, style 2 gages shall indicate the true pressure at the same temperature within plus 1 p.s.i. or minus 2 p.s.i. at 75 p.s.i.g. reading, and within plus or minus 3 p.s.i. at 100 p.s.i.g. reading. All type II, class B gages shall indicate the true pressure at minus 20° F. to minus 25° F. temperature within plus or minus 3 p.s.i. at 40 p.s.i.g. pressure. The style 4 gage shall indicate the true pressure within plus or minus 0.5 p.s.i. between the range of 4 p.s.i.g. through 16 p.s.i.g. at a temperature of plus 65 to 75° F. Gages shall indicate the true pressure at either of the above temperatures within the aforementioned limits within 5 seconds after the pressure has been applied.

3.4.3.4.2.2 Type II, class B, style 1, low pressure gage. Gages of this type, class, and style shall have a low pressure reading of 10 p.s.i.g or less, and a high pressure reading of not less than 45 p.s.i.g. The indicator bar shall be graduated in 1 p.s.i. increments, with the appropriate numerical value at each five-pound unit or less. These gages shall be similar in design to Figure 4.

3.4.3.4.2.3 Type II, class B, style 2, high pressure gage. Gages of this type, class, and style shall have a low pressure reading of 20 p.s.i.g. or less, and a high pressure reading of not less than 100 p.s.i.g. The indicator bar shall be graduated in 5 p.s.i. increments maximum, with the appropriate numerical value at each 10 p.s.i. interval. These gages shall be similar in design to Figure 4.

3.4.3.4.2.4 Type II, class B, style 3, air and liquid gage. Gages of this type, class, and style shall have a minimum pressure range of 12 p.s.i.g. to 45 p.s.i.g. The indicator bar shall be graduated in 1 p.s.i. increments, with the appropriate numerical value at each five-pound unit or less. The gage shall be fully protected against mud, sand, dust, brine, antifreeze solutions, and other harmful materials. The indicator bar shall be attached to the internal mechanism in such a manner as to facilitate flushing and oiling of the integral mechanism by pump action of the indicator bar. A hole shall be provided in the outer end of the barrel to permit injection and ejection of a liquid cleaner and light oil. These gages shall be similar in design to Figure 4A.

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3.4.3.4.2.5 Type II, class B, style 4, extra low pressure gage. Style 4 gages shall have a low pressure reading of 2 p.s.i.g. or less, and a high pressure reading of not less than 16 p.s.i.g. or more than 22 p.s.i.g. The indicator bar shall be graduated in increments of 1 p.s.i. or less on at least 2 sides, spaced a minimum of 0.0625 inch apart. Graduations shall be numerically marked each two pound unit or less. These gages shall be similar in design to Figure 4.

3.4.4 Types III and IV inflator gages. Types III and IV inflator gages shall consist essentially of a pneumatic tire pressure gage, a control valve, and an attached length of hose equipped for supplying compressed air to a valve stem by means of a trigger or lever mechanism. The gage shall indicate the air pressure upon release of the trigger or lever. The trigger or lever shall be located in a manner to facilitate the operation of the inflation attachment when the operator is wearing heavy gloves or mittens. The trigger or lever shall be hinged at, or near, one end and shall develop a leverage ratio of not less than 3-to-1 on the air valve assembly. All types III and IV gages shall be constructed to withstand a pressure of 175 p.s.i.g. without leakage. The gage shall have a low pressure reading of 10 p.s.i.g. or less and a high pressure reading of not less than 120 p.s.i.g., with graduations each 2 p.s.i. or less the entire length of the scale. Each 10 p.s.i. increment shall be clearly marked with the appropriate numerical value.

3.4.4.1 Accuracy. Accuracy requirements for types III and IV gages shall be as follows: Construction shall be such as to insure accuracy within the tolerance of plus or minus 0.5 of 1 percent of the full-scale reading at the 30 p.s.i. calibration, and within the tolerance of plus or minus 2 percent of the full-scale reading at the 80 p.s.i. calibration when subject to temperatures of from plus 110° F. to plus 115° F. At temperatures of minus 20° F., the accuracy shall be within the tolerance of plus or minus 1 percent of the full-scale reading at the 30 p.s.i. calibration, and within the tolerance of plus or minus 3 percent of the full-scale reading at the 80 p.s.i. calibration.

3.4.4.2 Discharge hose. Gages of types III and IV shall be provided with a 0.25-inch inside diameter (i.d.) discharge hose not less than 14 inches in length and in accordance with tire gage manufacturer's commercial practice. The hose shall be fitted on one end with a male threaded connector and on the other end with a full swivel connector.

3.4.4.3 Extension and foot. Gages of types III and IV, styles 2 and 3 shall be furnished with a metal extension not less than 5.5 nor more than 6.5 inches in overall length. The extension shall be fitted on one end with a female threaded connector and on the other end with a dual-chuck foot of the size as hereinafter defined for each type and style gage.

3.4.4.4 Couplings and connectors, quick-disconnect. Gages of types III and IV, style 3 shall be furnished with quick-disconnect couplings and connectors in accordance with MIL-C-4109. The couplings shall be class A, style 2. The connectors shall be class A, style 1.

3,4.4.5 Type III, internal indicating inflator gage. Gages of this type shall be of three styles and shall be internal indicating with maximum graduations of 2 p.s.i. increments. The internal indicating and air control assemblies shall be separate self-contained units (see 3.3), removable and replaceable without the use of special tools. Unless otherwise specified (see 6.2), the air valve element, gage cartridge, and sealing washers shall comply with Figure 5.

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3.4.4.5.1 Body. The body of type III gages shall be of cast aluminum alloy or similar metal and furnished with an air inlet, outlet, air screen, and valve actuating lever or trigger. The air inlet opening shall be threaded to receive a 0.25-inch National Pipe Thread (N.P.T.) male fitting. A wire-cloth air screen, 40 mesh, shall be furnished for the inlet opening and be readily accessible for cleaning. The valve actuating lever shall be hinged at, or near, one end and shall develop a leverage ratio of not less than 3-to-1 on the air valve assembly.

3.4.4.5.2 Type III, style 1, single-chuck foot gage. Gages of this type and style shall be furnished with a standard bore single-chuck foot.

3.4.4.5.3 Type III, style 2, dual-chuck foot gage. Gages of this type and style shall be furnished with a standard dual-chuck foot and shall be similar in design to Figure 6.

3.4.4.5.4 Type III, style 3, standard and large bore dual-chuck foot gage. Gages of this type and style shall be furnished with a standard bore dual-chuck foot and a large bore dual-chuck foot. Both the dual-footings shall be removable and replaceable at the end of the discharge hose by means of quick-disconnect fittings (see 3.4.4.4). Unless otherwise specified (see 6.2), each of the dual-footings shall be attached by a chain with rings to prevent the loss of either chuck while not in use. The ring shall be so constructed as to slide freely over the discharge hose, and shall be of such diameter as not to pass over the coupler. These gages shall be similar in design to Figure 6A.

3.4.4.6 Type IV, external indicating inflator gage. Gages of this type shall be of three styles and shall be external indicating with maximum graduations of 2 p.s.i. the entire length of the scale.

3.4.4.6.1 Body. The body of type IV gages shall be of cast aluminum alloy or similar metal and furnished with an air inlet with screen, outlet, and valve control lever or trigger. The air inlet opening shall be threaded to receive a 0.25-inch N.P.T. male fitting. A wire-cloth air screen, 40 mesh, shall be furnished for the inlet opening and be readily accessible for cleaning. The air outlet opening shall be threaded to receive the N.P.T. male fitting of the discharge hose.

3.4.4.6.2 Type IV, style 1, single-chuck foot gage. Gages of this type and style shall be furnished with a standard bore single-chuck foot.

3.4.4.6.3 Type IV, style 2, dual-chuck foot gage. Gages of this type and style shall be furnished with a standard bore dual-chuck foot and be similar in design to Figure 7.

3.4.4.6.4 Type IV, style 3, standard and large bore dual-chuck foot gage. Gages of this type and style shall be furnished with a standard bore dual-chuck foot and a large bore dual-chuck foot. Both dual footings shall be removable and replaceable at the end of the discharge hose by means of quick-disconnect fittings (see 3.4.4.4). Unless otherwise specified (see 6.2), each of the dual footings shall be attached by a chain with rings to prevent the loss of either chuck while not in use. The ring shall be so constructed as to slide freely over the discharge hose, and shall be of such diameter as not to pass over the coupler. These gages shall be similar in design to Figure 8.

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3.5 Storage temperatures. All gages and all components thereof shall be able to withstand without damage, storage in a temperature of minus 60° F. to 65° F. for a period of 72 hours, and a 24-hour cycle of temperature, starting at 10 hours at plus 70° F. to 90° F., followed by 5 hours of steady increase to plus 150° F. to plus 155° F., 4 hours at plus 150° F. to 155° F., and 5 hours of steady decrease to plus 70° F. to plus 90° F. Following these exposures, and after being stabilized at a temperature of plus 70° F. plus or minus 5° F., any gage of any type, class, or style shall be capable of meeting the applicable accuracy and performance requirements set forth in this specification.

3.6 Identification marking. Unless otherwise specified (see 6.2), type I and type II, class A gages shall be legibly marked with the manufacturer's name or trademark and the Federal Stock Number. Type II, class B gages shall be legibly marked with the manufacturer's name or trademark. Type III and type IV inflator gages shall be marked in accordance with MIL-STD-130.

3.7 Technical data. Data shall be furnished as specified (see 6.3).

3.8 Workmanship. The workmanship of the gage and accessory equipment shall be of the quality prevalent among manufacturers normally producing equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and and services conform to prescribed requirements.

4.2 Classification of inspection.

4.2.1 First article inspection. First article inspection applies to the pre-production model or initial production item noted in 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to each gage prior to being submitted for acceptance under the contract. The inspection shall consist of the examination in 4.3, accuracy tests in 4.4.1, and inspection of preparation for delivery in 4.5. Failure of the unit to pass examination or test shall be cause for rejection.

4.2.2.1 Sampling. When specified (see 6.2), sampling for lot acceptance shall be in accordance with MIL-STD-105 at the inspection level and acceptable quality level (AQL) specified in the contract. Samples for the tests will be selected at random from the samples that passed the examination in 4.3.

4.2.3 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the gages shall be examined and tested to determine conformance with the requirements of section 5 of this specification and as required by specifications referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with MIL-STD-105 and an AQL of 4.0 percent defective.

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4.3 Examination. Each production item or sample shall be examined to determine compliance with the requirements in section 3, using the following classification of defects:

Classification of defects

Major defects

(AQL shall not exceed 2.5 percent defective)

101. Defective casting or forging affecting operation (see 3.2.1).
102. Protective finish not as specified (see 3.3.9).
103. Chuck foot not as specified (see 3.4.2).
104. Dual-chuck foot not as specified (see 3.4.2.2).
105. Indicator bar not as specified (see 3.4.3.2).
106. Pressure range not as specified (see 3.4.3.3, 3.4.3.4.1, 3.4.3.4.2.2, 3.4.3.4.2.3, 3.4.3.4.2.4, and 3.4.4).
107. Indicator bar markings not as specified (see 3.4.3.2, 3.4.3.3, 3.4.3.4.1, 3.4.3.4.2.2, 3.4.3.4.2.3, 3.4.3.4.2.4, 3.4.4, 3.4.4.5, and 3.4.4.6).
108. Valve control lever or trigger inoperative (see 3.4.4).
109. Discharge hose not as specified (see 3.4.4.2).
110. Air screen not as specified (see 3.4.4.5.1).

Minor defects

(AQL shall not exceed 4.0 percent defective)

201. Fastening devices not as specified (see 3.3.5).
202. Gages with tool marks or rough finish (see 3.3.8).
203. Dimensions not as specified (see 3.4.3.3, 3.4.3.4.1.2, 3.4.3.4.1.3, and 3.4.3.4.2).
204. Identification marking illegible (see 3.6).

4.4 Tests.

4.4.1 Accuracy. Gages shall be tested at a temperature of plus 70° F. plus or minus 5° F., and shall indicate the true pressure within the limits of accuracy required of the particular type, class, and style of gage being tested in relation to the temperature referenced above. An air pressure measuring device having an overall accuracy of one-fourth of 1 percent of full-scale value shall be used as a criterion (see 3.4.3.3.1, 3.4.3.4.1.1, 3.4.3.4.2.1, and 3.4.4.1).

4.4.2 Construction.

4.4.2.1 Overload. Types I and II gages shall be subjected to an overload of 25 percent plus or minus 5 percent in excess of its rated capacity for a minimum of 10 minutes. Types III and IV inflator gages shall be subjected to a pressure of 175 p.s.i.g. for a minimum period of 10 minutes. The gage shall then be subjected to the applicable accuracy test as outlined in 4.4.1 (see 3.3).

4.4.2.2 Concentrated load. Type II, class A gages shall be subjected to a load test. Place the gage in a horizontal position upon 0.625-inch diameter test pins set 7 inches apart for the style 1 gage and 9 inches apart for the style 2 gage. Place another 0.625-inch diameter test pin at point contact centrally located equal distance between the test pins, on top of the gage barrel, then slowly apply a force up to 125 pounds. The gage shall then be examined for damage. This test shall not affect accuracy as specified in 4.4.1 (see 3.3).

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4.4.2.3 Indentation. Type II, class A gages shall be subjected to an indentation test. Place the gage barrel or stem on a flat surface, then place a 0.0625 inch radius test pin at any point on the barrel or stem and slowly apply a load up to 125 pounds. The gage barrel or stem shall then be examined for penetration. This test shall not affect accuracy as specified in 4.4.1 (see 3.3)

4.4.3 High and low temperature operational. Gages shall be tested at the temperature and air pressure specified in the applicable requirement for accuracy and shall indicate the true pressure within the limits of accuracy required for the particular type, class, and style of gage being tested. An air pressure measuring device having an overall accuracy of one-fourth of 1 percent of full-scale value shall be used as a criterion. All types I and II gages shall indicate the true pressure within the specified tolerances at the applicable temperatures and pressure within 5 seconds after pressure has been applied (see 3.4.3.3.1, 3.4.3.4.1.1, 3.4.3.4.2.1, and 3.4.4.1).

4.4.4 Storage.

4.4.4.1 Cold. The gage shall be placed in an ambient air temperature of minus 60° F. to minus 65° F. for a period of 72 hours. At the end of this period, the gage shall be allowed to stabilize at an air temperature of plus 70° F. plus or minus 5° F. The gage shall then be inspected for evidence of damage and tested in accordance with 4.4.1 (see 3.5).

4.4.4.2 Heat. The gage shall be subjected to a 24-hour cycle of temperature beginning with 10 hours at plus 70° F. to plus 90° F., followed by 5 hours of steady increase to plus 150° F. to plus 155° F., 4 hours at plus 150° F. to plus 155° F., and 5 hours of steady decrease to plus 70° F. to plus 90° F. The gage shall then be inspected for evidence of damage and tested in accordance with 4.4.1 (see 3.5).

4.4.5 Endurance. The types I and II gages shall be actuated by the application of air pressure and the indicator bar closed after each application of air, for not less than 2,000 gaging operations or cycles. The types III and IV gages shall be actuated by the application of air pressure and the air valve operated so as to return the gage to zero after each application of air, for not less than 10,000 gaging operations or cycles. The type I and II, class A gages shall be operated from an air source of 150 to 160 p.s.i.g. The type II, class B, styles 1 and 3 gages shall be operated from an air source of 45 to 50 p.s.i.g. The type II, class B, style 2 gage shall be operated from an air source of 90 to 100 p.s.i.g. The types III and IV gages shall be operated from an air source of 150 p.s.i.g. and the lever or trigger shall be actuated not less than 10,000 times to assimilate the required number of gaging operations. The gaging operations or cycles shall be at a rate of approximately 10 per minute for all types of gages. The method of applying air to the gages, actuating the indicator bar on the types I and II gages, and actuating the lever or trigger on the types III and IV gages is at the discretion of the testing facility. The gages shall then be examined for damage and the accuracy shall be tested as outlined in 4.4.1 (see 3.3.1).

4.4.6 Shock. The tire gage shall be dropped from a vertical distance of 3 feet onto a concrete surface five times for types I and II gages and 50 times for types III and IV gages. The gage shall be examined for damage and the accuracy shall be tested in accordance with 4.4.1 to determine compliance with 3.3.3. Failure of any gage to pass this test shall be cause for rejection.

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4.4.7 Flammability. Synthetic materials for test shall be extended outward and clamped 0.25-inch above a 5-inch square piece of 10-mesh Bunsen burner gauze in a horizontal position. A Bunsen burner with an efficient flame 0.5 to 0.75 inch in height shall be adjusted so that the flame tip will just contact the specimen when placed under the free end of the sample. At the end of 30 seconds, the flame shall be removed and the material allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the sample after the removal of the Bunsen burner. In case the material does not continue to burn after the first ignition, the burner shall be replaced under the free end for a period of 30 seconds immediately following the extinction of the flame. If the sample does not continue to burn until the flame has reached the end of the material after the second ignition, the sample shall be considered self-extinguishing. When tested as described, the synthetic materials shall not burn at a rate exceeding 2 inches per minute (see 3.2).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1. Cleaning and drying. The gages shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.1.1.2 Unit packaging. Type II gages shall be individually packaged in accordance with method IC-3 of MIL-P-116. Types I, III, and IV gages shall be individually packaged in accordance with method IC-2 of MIL-P-116.

5.1.1.3 Intermediate packaging. Gages of like description shall be intermediate packaged in a snug fitting fiberboard box conforming to PPP-B-636, class weather resistant. The box shall be closed in accordance with the appendix to the box specification. The gross weight and cubic displacement of the box shall not exceed 35 pounds and 1.5 cubic feet. (In the event gages are to be packed for level B as specified in 5.2.2, types III and IV gages require no intermediate packaging).

5.1.2 Level B. Unless otherwise specified (see 6.2), level B preservation and packaging shall be as follows (see 6.4).

5.1.2.1 Unit packaging. Each tire pressure gage shall be packaged in a paperboard box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. The box shall be closed in accordance with the applicable box specification to prevent accidental opening.

5.1.2.2 Intermediate packaging. Ten unit packages of tire pressure gages of like description shall be intermediate packaged in a paperboard or fiberboard box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636. The box shall be closed in accordance with the applicable box specification to prevent accidental opening.

5.1.3 Level C. The gages shall be packaged in accordance with the manufacturer's standard practice, providing that this insures protection for the gages during shipment and provides for safe delivery to their destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 Level A. Unless otherwise specified (see 6.2), packaged gages shall be packed in a cleated plywood box conforming to PPP-B-601, overseas type. Exterior shipping containers shall accommodate the greatest number of items in the least number of containers. Blocking, bracing, and cushioning shall conform to MIL-STD-1186.

5.2.2 Level B.

5.2.2.1 Military agencies. Unless otherwise specified (see 6.2), packaged gages shall be packed in a fiberboard box conforming to PPP-B-640, class 2. The gross weight of the container shall not exceed 200 pounds. Blocking, bracing, and cushioning shall conform to MIL-STD-1186. The box shall be closed in accordance with the appendix to the box specification.

5.2.2.2 Civil agencies. Five intermediate packaged gages of like description shall be packed in a close fitting fiberboard box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The gages, packaged as specified in 5.1.3, shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification rules and the National Motor Freight Classification rules. When fiberboard is utilized, it shall conform to PPP-F-320.

5.3 Marking. In addition to the marking specified in the procurement documents, the unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 for civil agencies, and MIL-STD-129 for Military activities.

6. NOTES

6.1 Intended use.

6.1.1 Type I. This precision gage is intended for use to check the accuracy of general service gages.

6.1.2 Type II, class A. This general service gage is intended for use in service areas to measure or check the pressure of air in tires or similar items.

6.1.3 Type II, class B. This general service gage is intended for use as a personal item for frequent measuring or checking the pressure of air in tires or similar items. It is intended to be carried in a person's pocket or storage compartment of a vehicle. It is not intended for rough usage or handling as often found in service areas.

6.1.4 Types III and IV. These inflator gages are intended for use on air supply lines to inflate, deflate, and measure or check the pressure of air in tires or similar items. They are intended to simplify the servicing of tires and similar items into one operation.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style required (see 1.2.1).
(c) First article, when required (see 3.1).
(d) Finish, if different (see 3.3.9).
(e) Chuck foot, if different (see 3.4.2).
(f) Gage overall length, if different (see 3.4.3.4.1.3).
(g) Internal mechanism, if different (see 3.4.4.5).

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(h) Chain with rings, if different (see 3.4.4.5.4, 3.4.4.6.4).
(i) Identification marking, if different (see 3.6).
(j) First article inspection, if different (see 4.2.1).
(k) Quality conformance inspection, if different (see 4.2.2).
(l) Sampling for lot acceptance, when required (see 4.2.2.1).
(m) Selection of applicable level of preservation, packaging, and packing required (see 5.1 and 5.2).
(n) Level B preservation and packaging, if different (see 5.1.2).
(o) Shipping containers, if different (see 5.2.1 and 5.2.2.1).

6.3 Contract data requirements. Required technical data, such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.7).

6.4 Level B preservation and packaging. Level B (see 5.1.2), has been developed for use by the Army and Air Force for civil agency procurement. For Navy procurements details of level B preservation and packaging, when required, should be developed at the time of procurement when all conditions are known.

MILITARY INTEREST:

Custodians:
Army - WC - AR OR AL / AR
Navy - YD
Air Force - 82

Review activities:
Army - ME, SM, WC
Navy - None
Air Force - None

User activities:
Army - AV, AT
Navy - MC
Air Force - None

Preparing activity:
Army - WC

CIVIL AGENCY INTEREST:
GSA-FSS

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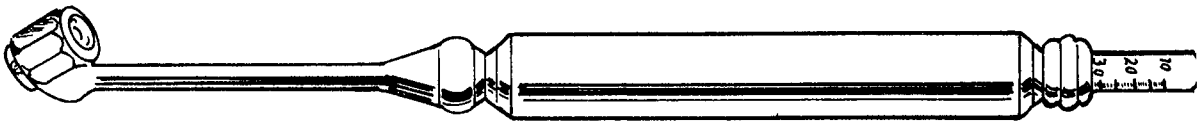


FIGURE 1. Type I precision gage.

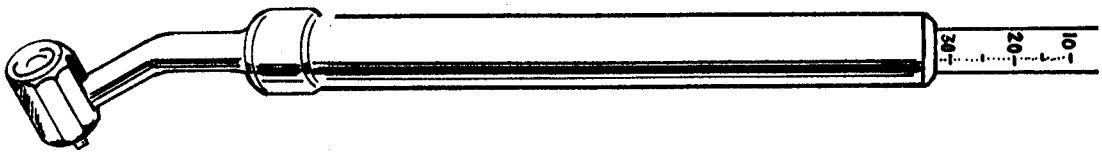


FIGURE 2. Type II, class A, style 1 general service gage.

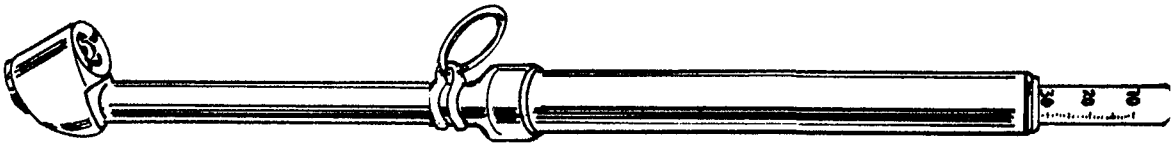


FIGURE 3. Type II, class A, style 2 general service gage.

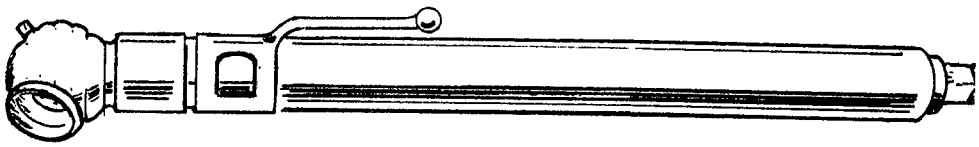


FIGURE 4. Type II, class B, style 1, 2, and 4 general service gage.

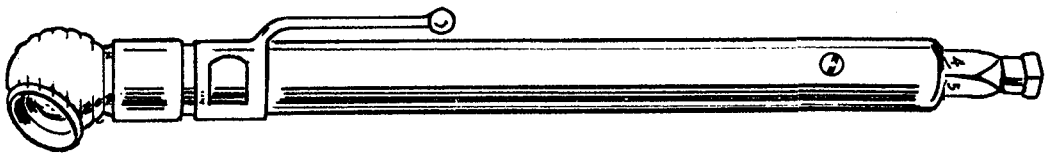


FIGURE 4A. Type II, class B, style 3 general service gage.

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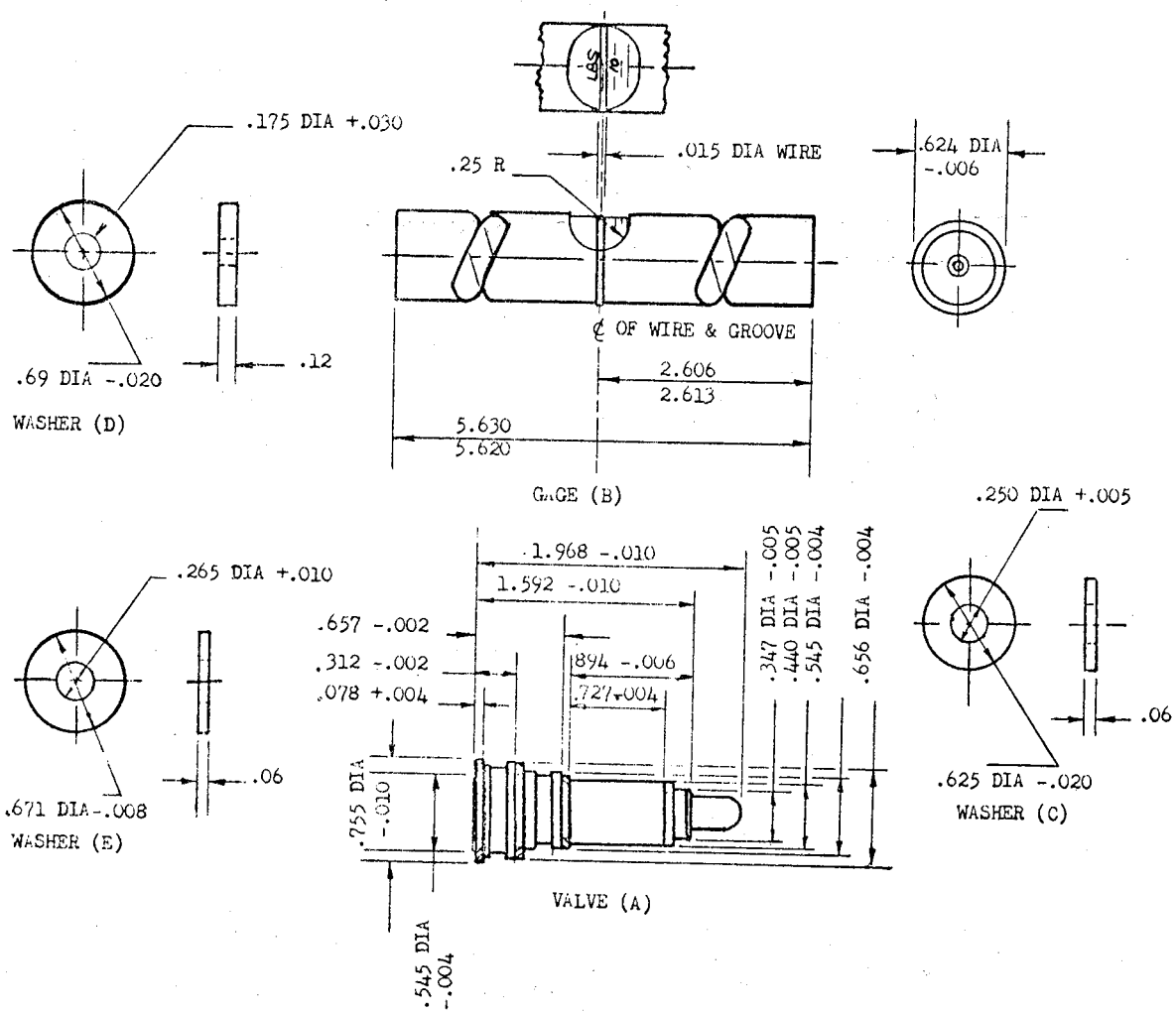


FIGURE 5. Gage cartridge, valve element & sealing washers for Type III gages.

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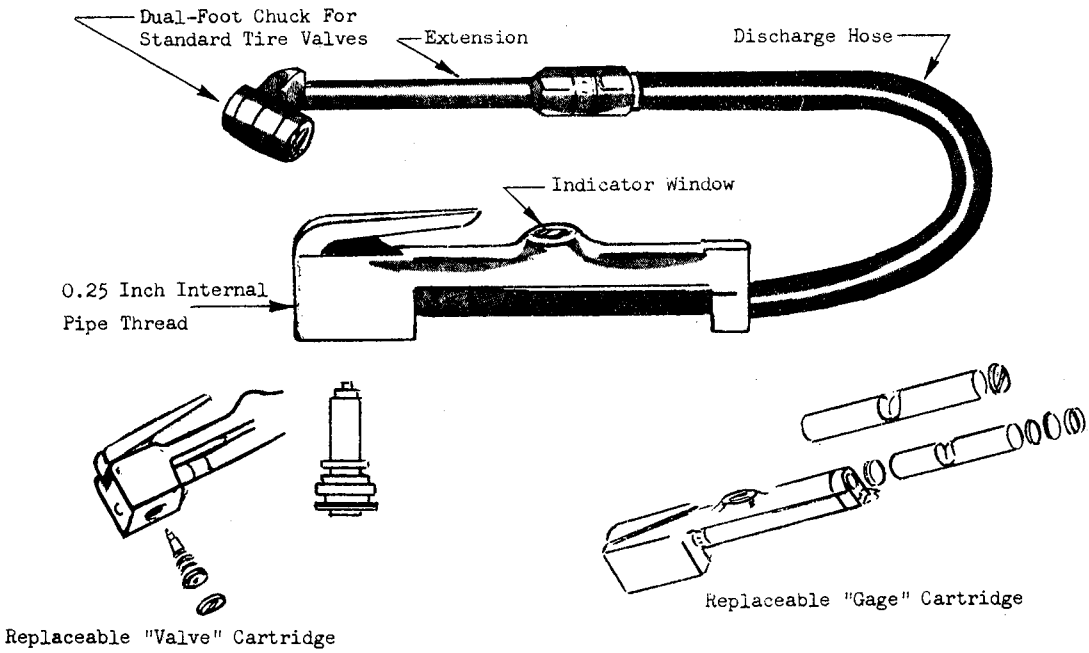


FIGURE 6. Type III, style 2 inflator gage.

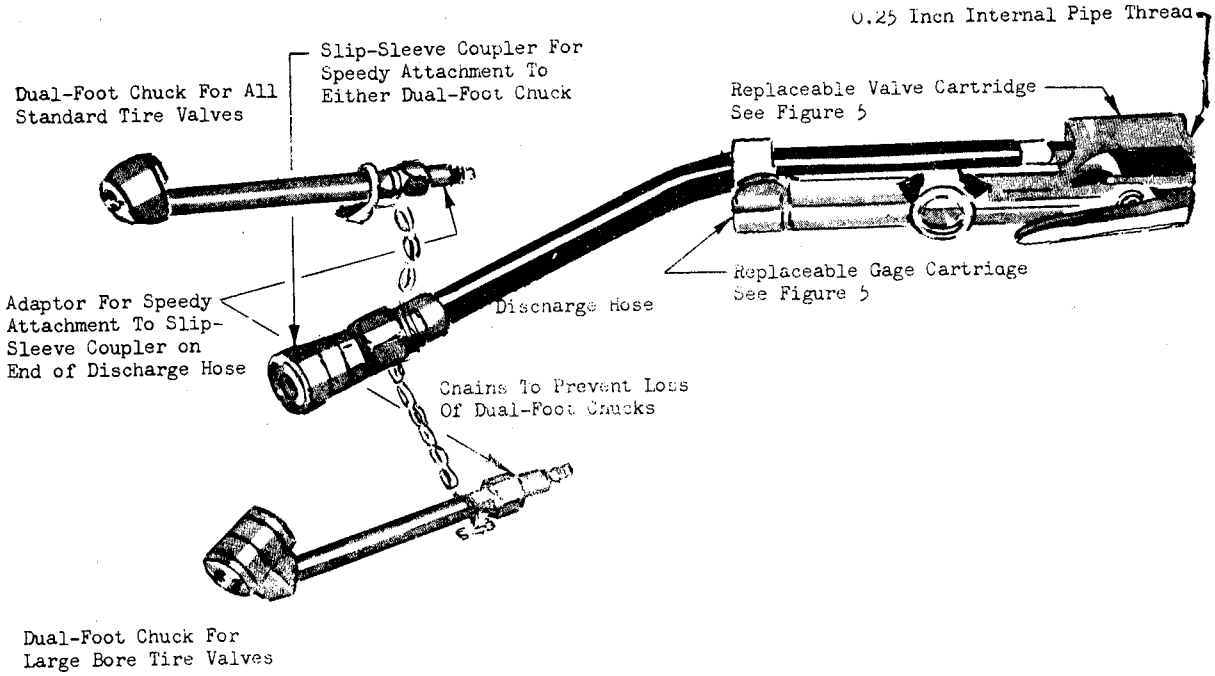


FIGURE 6A. Type III, style 3 inflator gage.

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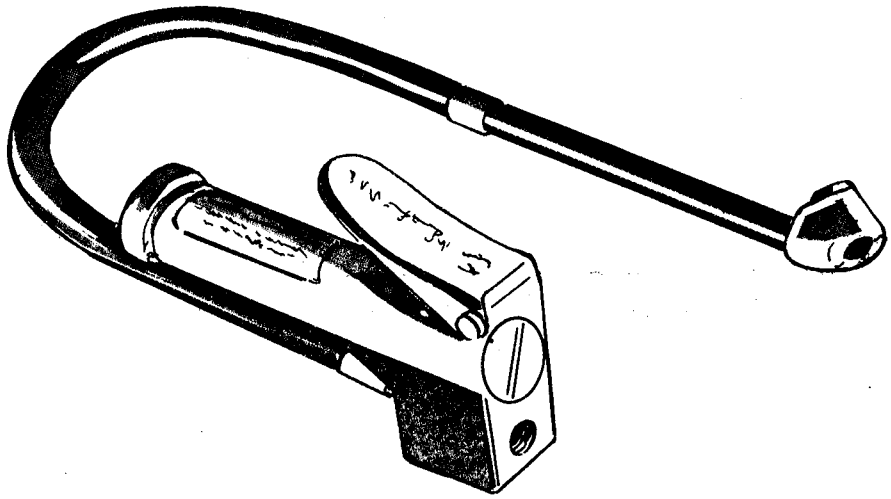


FIGURE 7. Type IV, style 2 inflator gage.

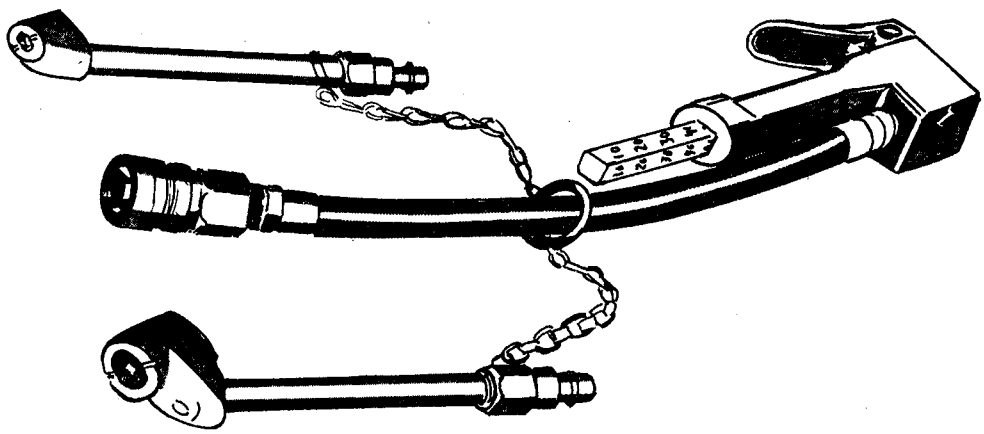


FIGURE 8. Type IV, style 3 inflator gage.

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